Mohamed Hossam

backend software engineer
Beni Suef, Egypt · Email · Phone · Portfolio · GitHub · LinkedIn · CodeForces · LeetCode

SUMMARY

Software Engineer with Python, Django, and FastAPI. Builds complete systems end-to-end (schema, services, auth, deployment) with emphasis on security and data integrity. Competitive programming strengthens edge-case reasoning, correctness and performance. Seeking a junior/intern role to find and solve more challenging problems in production.

TECHNICAL SKILLS

Backend & APIs: Django, DRF, FastAPI, REST APIs, Celery, Async, WebSockets, Pydantic

Software Design & Quality: API Design, Service-Layer Architecture, Automated Testing

Databases: PostgreSQL, MySQL, Redis, SQLAlchemy, Database Indexing & Query Optimization.

Infra & Deployment: Docker, Compose, Nginx, Gunicorn, Uvicorn, Git, CI/CD

Languages: Python, SQL, JavaScript, C++, Bash

OPEN SOURCE CONTRIBUTIONS

NetBox — Merged PR #23378 *Oct 2026*

- Eliminated a background-job flood in NetBox's global search replaced the one-RQ-job-per-write path with a cross-transaction Redis coalescing buffer that merges updates across 98 searchable models into a single drain job per 2-second window collapsing a reporter's ~237k-job backlog and keeping indexing eventually consistent at any write throughput.
- Made the coalesced pipeline crash-safe and duplicate-free: an atomic Redis RENAME claim plus a PostgreSQL session-level advisory lock spanning apply-and-cleanup stop concurrent appliers from duplicating CachedValue rows under READ COMMITTED (no unique constraint), while a 5-minute safety-net job idempotently replays stranded batches and a retry cap discards poison batches with rolling-upgrade back-compat and 67 tests, including parallel-suite isolation.

NetBox — Merged PR #22834 *Jul 2026*

- Enforced context-aware HTML sanitization: restricted img[src] to HTTP(S)/relative via nh3 + regex; blocked ssh:/file:/ftp: bypasses across 3 rendering paths.
- Shipped 7 regression tests covering whitespace/control-char evasion, javascript: blocking, and scheme-intersection enforcement.

Django CRM — Merged PR #449 #456 *Jun–Aug 2026*

- Added Product.main_image: ImageField with migration, lifecycle cleanup, auto-resize tro 200×200 PNG, admin preview/thumbnail.
- Fixed TransactionQualityEvent admin UX: hid attachment icon when empty and added stacked inline with XSS-safe file links.

TECHNICAL PROJECTS

YouTube Clone *Django 6, PostgreSQL 16, Redis 7, Celery 5, ClamAV, Docker, ImageKit CDN* *2026 – Present*

- Architected 3-layer platform with zero-byte app storage; served via signed ImageKit CDN URLs for adaptive HLS (240p–1080p) and watermarked thumbnails.
- Engineered a defense-in-depth upload: magic header sniffing, ffprobe validation, SHA-256 dedup to reject malformed or duplicate files before quarantine.
- Asynchronous malware scanning with Celery and ClamAV behind a quarantine step, ensuring files remain non-streamable until cleared; implemented retries with backoff for infrastructure failures,

with infected files rejected and never exposed to viewers.

— Concurrency-safe abuse controls: Redis token-bucket per IP/user; 24h view dedup with atomic F() increments and double-check transactions to prevent redundant writes under concurrent load.

FastAPI OCR Microservice *FastAPI, Tesseract, Docker, Pydantic v2, Prometheus* *2026 – Present*

— Characterized capacity at 7.2 req/s and 890ms p95 under 50 concurrent requests, correlating throughput and tail latency with CPU and queue depth via Prometheus metrics and structured logging.

— Prevented overload by bounding CPU-bound Tesseract behind worker pool and queue backpressure with 429, selecting limits from measured throughput plateau and p99 tradeoffs.

— Hardened stateless FastAPI service for production with 10MB upload limit, 30s timeout, consistent error handling, health/readiness probes, and Docker with automated tests in CI.

— Reduced OCR error rate by 62% (CER 8.4% → 3.1%) via ablation-tested Pillow preprocessing, keeping only stages with proven accuracy gain vs latency cost.

LEADERSHIP & COMMUNITY

Leader, Mentor & Problem Setter *ICPC Nahda University Community* *2025 – Present*

— Mentored 600+ competitive programmers, helping them turn data-structures and algorithms knowledge into practical problem-solving instincts.

— Authored original problems for internal training contests/practice **sheets** aligned with ICPC standards.

ACHIEVEMENTS & CERTIFICATIONS

— GitHub: Ranked #7 in Egypt by activity (**Committers.top**)

— QECPC 2025: Honorable Mention • ECPC Finalist 2026

— Constructor Open Cup 2026: Ranked 20th internationally (Constructor University & JetBrains Foundation)

— Codeforces Candidate Mater rating (top 45 in Egypt); solved 2,400+ algorithmic problems across Codeforces, LeetCode, and vJudge

— Certifications: ALX Professional Foundations; ALX Freelance Academy

EDUCATION

B.Sc. in Computer Science Nahda University, Beni Suef Expected 2028
• GPA 3.26/4.0
Relevant Coursework: Algorithms, (OOP), Database, Operating Systems, Computer Networks